Scenario 2: Pairing historical streamflow data with flood maps to identify vulnerable areas

Description

You are part of a watershed alliance working on a community resilience grant. Your plan includes an analysis of historical flood exposure and inundation risks, with the intention of prioritizing vulnerable communities.

Your Task

- Select your region from the HOME page or from the dropdown on the MAPS page.
- Click the icon with three stacked layers (at the top right of the map) to open the map layers menu. From there, select your preferred basemap. For example, 'USGS Imagery' displays high-resolution satellite imagery, which is useful for visually identifying landmarks and natural features, while 'ESRI World Imagery' offers detailed topographic information that helps in interpreting terrain in flood-prone areas. 'CartoDB Positron' provides a cleaner, less detailed background, ideal for users who prefer a minimal map to focus solely on the flood extent.
- Similar to Scenario 1, zoom in to find a river of interest and click the river to select it. Then, use the 'Historical' data feature to identify what the highest estimated streamflow value was between 2018 and the present.
- If flood maps are available for the selected river, a vertical slider will appear on the right side of the page. This slider allows you to explore flood scenarios under different streamflow conditions. As you move the slider, the corresponding pre-computed flood extents will display on the map, indicating areas of possible inundation. Move the slider so it displays the inundation extent of the highest estimated streamflow value on the historical data graph.

  **Note: These maps are pre-generated for a set of representative conditions, so you may not be able to display the exact streamflow or stage value you are interested in. However, by navigating to the closest value you can get an indication of what the inundation extent may look like.*
- Consider social vulnerability layers (if your region has any within the layers panel), or assess what types of infrastructure might be at risk, in areas that consistently show up as flooded. You may want to adjust the base maps and the transparency of the flood map layer using the X function.
- Optional: For an additional source of inundation maps, consider visiting the [USGS Flood Inundation Mapper] website. Please note that USGS only provides this service for select gauged locations.
- Evaluate whether FloodSavvy and additional data sources provide sufficient detail to identify areas vulnerable to inundation under high streamflow conditions.